

Downloaded samplesgshA1.txt file

- 1.) Check quality with fastqc, but using falco (Galaxy Version 1.3.0+galaxy0) b/c faster
 - Default parameters
 - Run with transcriptome reads
 - Forward data
 - Fails: per tile sequence quality; Sequence duplication levels; Adapter Content
 - Warnings: Per base sequence content; Per sequence GC content; Overrepresented sequences
 - Reverse Data
 - Fails: Per tile sequence quality; adapter content
 - Warnings: Per base sequence content; Per sequence GC content; Sequence duplication levels; Overrepresented sequences

- 2.) Trim with fastp (Galaxy Version 1.3.3+galaxy0)
 - Set to paired collection
 - Gets rid of stuff that are fails for the quality control
 - Observe the results on HTML, those under filtering reports

- 3.) HISAT2 (Galaxy Version 2.2.2+galaxy0) - aligns transcriptome reads to the genome that is provided by Dr. Van Laar
 - Change in parameters:
 - Source for reference genome: genome from the history
 - Single or paired library? Set to paired-end dataset collection

Made an alignment file called BAM

Now, what actual transcripts do we have?

Step4.) StringTie ((Galaxy Version 2.2.3+galaxy0) - assembles short reads into actual transcripts, predicts full length transcripts

- Make sure input option parameter is set to Short Reads

Created 6 files of transcripts

- Change parameter to reference genome to history
 - Burkholderia.gff

Step 4.) StringTieMerge (Galaxy Version 2.2.3+galaxy0) - instead of 6 files, puts them all together

- Use dataset collection and use the StringTie file
- Use reference as burkholderia.gff

- step 5.) featureCounts (Galaxy Version 2.1.1+galaxy0)

- Set alignment file to dataset collection and use HISAT file
- Set gene annotation file as GFF/GTF from history and use latest stringtiemerge
- Rewrite GFF gene identifier as transcript_id

- Does the input have read pairs? Set to Yes, paired-end and count them as 1 single fragment
- Next step: Deseq2 (Galaxy Version 2.11.40.8+galaxy2) - function? Takes all 6 conditions and sees if it is more represented in one condition or the other, gives statistical significance
 - Make sure to rename to Counts under featurecounts for easier setup
- Factor: set to Gene_Type
- 2 factor levels. Factor level 1 set as WildType; Factor level 2 set as Mutant
 - Rename the counts as wildtype or mutants
 - Tells if gene was up or down regulated
 - Under Advanced options:
 - Use beta priors? Set to yes
 - Under Output options: generate plots and output normalised counts
 - Click “include hidden” within history and locate the featurecount and find the Counts, rename as plate/BrothReplicate#
 - Put these in using the multiple dataset options, search with the ellipses
- From the result file you get a p-value, can determine null hypothesis
 - For the gene, there is no change in gene expression between wildtype and mutant: null hypothesis
 - If there are changes between the two, you can infer significant difference, but need this p-value to make a statistical backing for your reasoning
 - Less than 0.05, it is statistically significant

downloaded both the plot and the result file

Step 6.) filter data on any column using simple expressions

- Use on DEseq2 file
- With following condition - $c7 < 0.05$
 - Column 7 is the p-adj, run it on this one instead of the raw p-value column
- Run again with the following condition - $\text{abs}(c3) > 1$
 - Column 3 is the $\log_2(\text{FC})$, run on this
 - Only do this after the previous filtering is finished
- From the plots we download here, we learn about whether expression change between the two growth media/conditions is statistically significant enough to reject the null hypothesis

Step 7.) filter on dataset with following condition:

- $\text{abs}(c3) > 1$
 - Cut it down to 626 lines from 1010 lines

Step 8.) were gonna add a header now through a new Excel file

- Put the following in: GeneID Base mean $\log_2(\text{FC})$ StdErr Wald-Stats P-value P-adj

Step 9.) concatenate tool

- Insert the most recent filter on dataset file

Now our data file has proper headers!

- Rename the file to Differential Gene Expression of Mutant and Wildtype
- Make sure to save this as a csv file!

- Go to edit attributes > datatypes > target datatype: tabular (using 'Convert CSV to Tabular')

Step 10.) volcano plot (Galaxy Version 4.0.3+galaxy0) - represents p-values

- Change the respective columns into the following: 7, 6, 3, 1
- Downloaded this

Step 11.) opened the tabular broth vs plate file as an excel file

- Sort the entire document from smallest to largest in log2(FC), kept the top 10 and bottom 10, delete everything in between, then save as a .csv file (separately so that you can still have the initial file)
 - CTRL a > data > sort > input conditions log2(FC) and smallest to largest
 - Make sure you delete all the cells that aren't included in the list

Step 12.) upload that new.csv file to galaxy

- Edit attribute on galaxy > datatypes > make sure New Type is set as "tabular"

Step 13.) join 2 datasets side by side on a specified field (Galaxy Version 2.1.3)

- Join .csv file with normalized counts dataset
- Parameter change: Keep Header Lines - Yes

Step 14.) cut columns from a table (Galaxy Version 1.0.2)

- parameter : cut c8-c14
 - *may need to set as c9-c14 if there is a blank column, check by clicking on the file in galaxy and scrolling to the right to see if there is an additional column*
 - *didn't need to make adjustments, c8-c14 was fine*

Step 15.) heatmap2 - (Galaxy Version 3.3.0+galaxy0)

- Change parameters:
 - Data transformation - Log2(value+1) transform my data
 - Blue (high level expression) Wildtype has expression in the top half of the genes, and the same goes for the mutants
 - Brown (low level of expression) Wildtype has lower levels of expression in the bottom half of the genes, and the same goes for the mutants
 - Will download this file as well

Step 16.) filter data on any column using simple expressions (Galaxy Version 1.1.1)

- Filter - burkholderia.gff
- Wanna keep all under CDS under column 3
 - With following condition: c3=='CDS'

Now we wanna keep column 9

Step 17.) Cut column from a table (Galaxy Version 1.0.2)

- Retain only column 9
- Parameters: cut columns - c9

Cant read these intuitively so...

Step 17.) convert delimiters to TAB [Galaxy Version 1.0.1](#))

- Parameters:
 - Convert all - semicolons

We just want the gene name on column 2

Step 18.) replace text in a specific column [\(Galaxy Version 9.5+galaxy3\)](#)

- Parameters:
 - In column 2
 - Find pattern - Parent=
- Can preview under visualize for confirmation
 - Yup, confirmed it was taken out

Now we're interested in column 8 because it has the gene function
We want the gene name column 2 and the gene products column 8

Step 19.) cut columns from a table [\(Galaxy Version 1.0.2\)](#)

- Parameters:
 - Cut columns: c2,c8

Step 20.) join two datasets side by side on a specified field [\(Galaxy Version 2.1.3\)](#)

- Join cut on dataset with latest concatenate.csv file (the one you set as tabular)
- Keep header lines - yes